

SANA ULLAH

Software Engineering Student • AI & Machine Learning • Robotics Enthusiast

Malakand, Pakistan • +92-316-5610591 • sanaullah.research@gmail.com • linkedin.com/in/sana-ullah-960a50322

PROFESSIONAL SUMMARY

Motivated Software Engineering student with hands-on experience building and deploying end-to-end AI/ML systems, computer vision pipelines, and intelligent automation workflows using n8n. Passionate about advancing research at the intersection of Machine Learning, Robotics, and Autonomous Systems. Seeking to contribute to high-impact AI-driven projects while pursuing graduate-level research in Human-Robot Interaction, Computer Vision, and Emotion-Aware AI.

EDUCATION

Bachelor of Science in Software Engineering

2023 – Present

University of Malakand, Malakand, Pakistan

Relevant Coursework: Data Structures & Algorithms • Object-Oriented Programming • Database Systems • Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, HTML, CSS

AI / Machine Learning: Machine Learning, Deep Learning, Data Analysis, Model Deployment

Libraries & Frameworks: TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, Streamlit

Automation & Tools: n8n, API Integration, Workflow Automation, REST APIs

Core Competencies: Computer Vision, End-to-End ML Pipelines, AI Deployment, Automation Systems

PROJECTS

Plant Disease Detection System

2024 – Present

Convolutional Neural Network • Streamlit Deployment • End-to-End ML Pipeline

- Designed and trained a **CNN achieving ~96% validation accuracy** for multi-class plant disease classification, demonstrating production-grade model performance.
- Engineered a robust preprocessing and real-time **image augmentation pipeline** (rotation, flipping, brightness normalisation) to improve generalisation and reduce overfitting.
- Deployed an interactive **Streamlit web application** enabling real-time predictions from live image uploads with confidence-score visualisation, accessible to non-technical end users.
- Architected a fully reproducible **end-to-end ML pipeline** covering data ingestion, feature extraction, training, evaluation, and production deployment.

Real-Time Face Tracking System

2024

OpenCV • Computer Vision • Robotics Integration

- Developed a real-time face detection and tracking module using **OpenCV** (Haar Cascade + DNN detectors), achieving stable multi-frame tracking under variable lighting.
- Integrated the computer vision pipeline with a **robotics control interface**, demonstrating applied CV in embedded and autonomous systems contexts.

- Optimised frame processing latency for **edge hardware deployment**, establishing a foundation for future Human-Robot Interaction research.

Intelligent Workflow Automation Suite

2024

n8n · API Integration · AI-Augmented Automation

- Architected and deployed multiple **n8n automation workflows**, integrating REST APIs, webhooks, and third-party services to eliminate repetitive manual processes.
- Built **AI-augmented pipelines** combining LLM API calls with conditional logic nodes, automating multi-step data processing sequences for measurable efficiency gains.
- Designed modular, reusable **workflow templates** for API aggregation, scheduled reporting, and cross-service notification routing.

RESEARCH INTERESTS

Artificial Intelligence · Machine Learning · Human-Robot Interaction · Computer Vision · Autonomous Systems · Emotion-Aware AI

ADDITIONAL INFORMATION

Professional Skills: Analytical Thinking · Problem Solving · Research Mindset · Team Collaboration · Time Management

Languages: English (Professional) · Urdu (Fluent) · Pashto (Native)